CampusHub API Testing Lab Report

Course Controller JUnit Testing

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Test Results

Test ID	Expected result	Actual result	Pass / Fail
01 Find existing course	Calling `getCourseById(101L)` returns 200 OK. The course has ID 101 and code IS-401.	Returned 200 OK. The course ID was 101 and its code was IS-401.	**PASS**
02 Look up missing course	Calling `getCourseById(9999L)` returns 404 Not Found with an empty body.	Returned 404 Not Found with an empty body.	**PASS**
03 Create valid course	Creating IS-404, Software Testing, 3 credits returns 201 Created. The returned course has code IS-404 and 3 credits.	Returned 201 Created. The course code was IS-404 and the credit value was 3.	**PASS**
04 Reject zero credits	Creating IS-404, Software Testing, 0 credits returns 400 Bad Request and states that credit must be positive.	Returned 400 Bad Request. The message stated that a positive credit value is required.	**PASS**
05 Delete existing course	Calling `deleteCourse(103L)` returns 204 No Content. A later lookup of course 103 returns 404.	Returned 204 No Content. The later lookup returned 404 Not Found.	**PASS**

Final result: 5 tests run, 0 failures, 0 errors, and 0 skipped tests.

Maven Test Summary Screenshot

```
[INFO] -------------------------------------------------------
[INFO]  T E S T S
[INFO] -------------------------------------------------------
[INFO] Running edu.ku.campushub_api.controller.CourseControllerTest
[INFO] Tests run: 5, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.117 s

[INFO] Results:

[INFO] Tests run: 5, Failures: 0, Errors: 0, Skipped: 0

[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
[INFO] Finished at: 2026-11-04T23:25:21+04:30
[INFO] ------------------------------------------------------------------------
```

The tests were run with Maven using `CourseControllerTest`.